

■ Accounts Payable (AP) / Vendor Module & Business Cycles

D365FO Middleware Backend — Vendor AP Domain, Custody Filtering & Posting Flows

Target Audience: Finance (AP/Disbursements), Subcontractor Management, Software Engineers.

1. Business Domain Overview

The **Accounts Payable (AP)** module manages vendor expenses and subcontractor payments:

- **Vendor Freight Invoices:** Sea shipping line freight bills, airline cargo bills, and co-loader invoices.
- **Vendor Fleet / Trucking Invoices:** Bills from third-party hired truck owners, transport subcontractors, and fleet suppliers.
- **Vendor Adjustments & Credit Notes:** Cost adjustments, demurrage settlements, and rebate credits.
- **Vendor Payments:** Payment disbursements, withholding tax retention, and bank transfer matching.

2. AP Entry Processors & Business Scope

Processor Class Name	EntryProcessorTypes	D365FO Target Entity	Real-World Business Context
VendorFreightEntryProcessor	VendorFreight	LedgerJournalHeaders / Lines	Vendor Freight Bills (LedgerJournalTable)
VendorFreightAdjustmentEntryProcessor	VendorFreightAdjustment	LedgerJournalHeaders / Lines	Freight Vendor Invoice Cost Adjustments
VendorTruckingEntryProcessor	VendorTrucking	LedgerJournalHeaders / Lines	Hired Truck Owner & Subcontractor Bills
VendorTruckingAdjustmentEntryProcessor	VendorTruckingAdjustment	LedgerJournalHeaders / Lines	Fleet Vendor Adjustments
VendorPaymentFreightEntryProcessor	VendorPaymentFreight	Vendor Payment Journal	Freight Vendor Payment Disbursements
VendorPaymentTruckingEntryProcessor	VendorPaymentTrucking	Vendor Payment Journal	Fleet Subcontractor Payment Execution

3. Critical Business Rules & Accounting Impact

■ 1. Custody Account Exclusion Rule

In logistics, cash advances are often given to internal drivers or logistics coordinators for trip expenses. These internal advance accounts belong to the **Custody Vendor Group**.

- **Rule:** The vendor processors automatically scan lines. Any vendor account belonging to the **Custody** group is **EXCLUDED** from vendor bill posting (except for pure Ledger lines).
- **Reason:** Internal driver custody advances must NOT be booked as external vendor supplier liabilities—they are handled separately under Custody Settlement.

■ 2. Withholding Tax (WHT) Deduction

- When processing vendor invoices, the middleware checks if **ISWITHHOLDINGCALCULATIONENABLED** is true.
- It attaches **ITEMWITHHOLDINGTAXGROUPCODE** (e.g. **WHT_1%** or **WHT_3%**). Upon posting in D365FO, tax is deducted at source from the net payment payable to the vendor.

■ Accounting Posting Entries (Vendor Invoice Journal):

- **DEBIT:** Cost / Expense Main Account (e.g., **510101** Ocean Freight Expense or **510201** Trucking Subcontractor Expense)
- **CREDIT:** Vendor Subledger Account (**VendAccount** e.g. **VEND-00412**)
- **CREDIT:** Withholding Tax Payable (Deducted at source)

4. End-to-End AP Business Cycle Flow

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[Vendor Supplier Invoices & Subcontractor Trip Sheets]
■
▼
[API Endpoint: Upload Vendor AP Excel]
■
▼
[VendorFreightEntryProcessor]
(Applies Custody Filter, Looks up Vendor Tax ID)
■
▼
[DataBatch Status: PENDING (Max 1000 Lines)]
■
▼
[BullMQ Queue: dfo-vendor-journal-queue]
■
▼
[VendorJournalPostingStrategy]
■
▼
[Post OData to D365FO LedgerJournalHeaders / Lines]
■
▼
[D365FO Creates Posted Vendor Invoice Journal]
(Generates Journal Batch Number & Credits Vendor Subledger)
■
▼
[Batch Status Set to COMPLETED]

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Detailed Step-by-Step Cycle Execution:

1. **Vendor Sheet Upload:** AP staff uploads vendor bills including vendor account numbers, supplier invoice numbers, tax IDs, due dates, and trip references.
2. **Custody Scanning:** Lines are checked against Custody vendor master data; any internal custody vendor lines are filtered out to prevent accounting distortion.
3. **Vendor Enrichment:** Master data service retrieves vendor tax registration numbers, payment terms, and posting profiles from D365FO cache.
4. **Voucher Grouping:** Lines sharing the same **VOUCHER** and **INVOICE** number are grouped into a single D365FO journal header.
5. **Queue & OData Posting:** Job is pushed to **dfo-vendor-journal-queue**. The strategy creates a **LedgerJournalHeaders** record and populates **LedgerJournalLines** with **ACCOUNTTYPE = 'Vend'**.
6. **Finalization:** D365FO confirms journal creation, recording batch numbers in MongoDB.